

DHEERAJ REDDY REDDYGARI

+1 (716) 907 0001 | dheerajreddy5b7@gmail.com | Buffalo, NY, USA | LinkedIn

EDUCATION

University at Buffalo – SUNY, Buffalo, New York

01/2024 - 01/2025

Master's, Data Science

GPA: 3.52

Relevant Coursework: Machine Learning, Data Intensive Computing, Statistical Learning and Data Mining, Data Models and Query Languages, Storytelling through Data Visualization (Tableau), Data Structures and Algorithms.

G Pulla Reddy Engineering College, Kurnool, India

08/2019 - 05/2023

Bachelor's, Computer Science and Engineering

GPA: 8.07

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS)

Hyderabad, Telangana, India

Data Engineer Intern

03/2023 - 07/2023

- Developed and optimized workflows to process and integrate large datasets from multiple sources, boosted data integration accuracy by 25%.
- Enhanced data processing speed and reliability by 30% using Apache Spark and Hadoop for distributed computing.
- Managed and stored data securely on AWS, ensuring scalability and adherence to industry standards, with a 20% reduction in storage costs.
- Developed dashboards and visual reports in Tableau, providing actionable insights that improved decision making efficiency by 35%.

ACADEMIC PROJECTS

Analyzing and Predicting Traffic Crash Severity, Data Intensive Computing

08/2024 - 12/2024

- Led a team of 4 to develop a machine learning pipeline for predicting traffic crash severity using factors like weather, traffic, road conditions, injuries, and damages.
- Trained and evaluated 6 models, including logistic regression, random forest, XGB classifier, SVM, decision trees, and Naive Bayes.
- Leveraged Spark in distributed data processing to efficiently model large datasets.
- Built models using standard libraries and Spark's MLlib, evaluating their performance based on accuracy and training time.
- Achieved a 15–20% reduction in training time while maintaining model accuracy.

Home Credit Default Risk Database, Data Models and Query Languages

08/2024 - 12/2024

- Engineered a relational database from CSVs with 6 normalized tables, achieving BCNF compliance and enforcing data integrity.
- Streamlined SQL queries by implementing indexing, reducing execution time by 33% (from 134ms to 90ms) and eliminating 1,000+ duplicate records.
- Optimized SQL queries (JOIN, GROUP BY, subqueries) to analyze credit risk and identify loan applicant risk factors, reducing query execution time by 33%.
- Designed an E/R diagram and schema to process large datasets efficiently and created a website utilizing the database.

The Why and Where of Crime, A Data-Driven Approach to Prevention Strategies

01/2024 - 05/2024

- Conducted in-depth geospatial analysis on crime data, focusing on 200+ neighborhoods, which provided a detailed understanding of crime hotspots and influenced the reallocation of patrol resources to enhance community safety.
- Designed and validated 3 machine learning models, including Random Forest, SVM, and KNN, to predict crime patterns.
- Finalized and deployed a model with 92% accuracy using Streamlit, enabling real-time crime prediction and supporting decision-making.

SKILLS

- Programming Languages: C/C++, Java, Python, SQL, R
- Data Structures and Algorithms, Machine Learning, Neural Networks
- Statistical Learning, Data mining, Data Cleaning, Data Analysis, EDA, Data Warehousing
- Data Engineering: Data Pipelines, ETL, Cloud Infrastructure (AWS, Azure)
- AWS: SageMaker, S3, EC2, EMR, RDS (MySQL, PostgreSQL), DynamoDB
- Big Data Technologies: Apache Spark, Hadoop, Map Reduce, Hive, MLlib.

CERTIFICATIONS AND ACHIEVEMENTS

- AWS Certified Machine Learning Engineer – Associate.**
- Microsoft Certified:** Azure Data Fundamentals.
- AWS Cloud Computing** Course by Amazon Web Services.
- Achieved **1407th** rank in the global coding competition of TCS CodeVita Season 10.
- Solved **150+** problems in LeetCode.